The Normative Calculation Problem Under Heterogeneous Intentionality: Why Effective Enforcement Requires Price Mechanisms

Ignacio Adrián Lerer
Independent Researcher
Buenos Aires, Argentina
adrian@lerer.com.ar
ORCID: 0009-0007-6378-9749

December 2025

ABSTRACT

Legal systems invest massive resources in enforcement, yet corporate compliance remains low and recidivism high. I argue this puzzle is not solvable by increasing sanctions or enforcement resources. The problem is architectural: enforcement mechanisms designed for human moral agents (Level 3 intentionality) fail when applied to optimization algorithms embodied in corporate form (Level 1 intentionality). This paper connects two previously separate literatures: Asymmetric Intentionality Theory, which diagnosed the structural mismatch between legal design and corporate cognition, and the Normative Calculation Problem, which explains why legal systems fail to generate the information necessary for rational compliance calculation. I argue that enforcement mechanisms requiring common knowledge, shame, or reciprocity systematically fail against Level 1 entities. The solution is not more enforcement of the same type, but enforcement mechanisms that operate on verifiable outputs rather than inferred intentions. I examine algorithmic enforcement as a directional solution, document its limitations (Goodhart's Law, capture risk, rigidity), and propose hybrid mechanisms combining algorithmic verification with human oversight for ambiguous cases. The analysis has implications for mechanism design, AI governance, and legal reform in contexts where heterogeneous intentionality is unavoidable.

Keywords: heterogeneous intentionality, normative calculation, legal prices, enforcement mechanisms, asymmetric intentionality, Mises, Hayek, corporate compliance, algorithmic enforcement, institutional economics

JEL Codes: K00, K12, K42, D02, D80, B25, P48

I. INTRODUCTION: THE ENFORCEMENT PUZZLE

I.A. The Empirical Puzzle

Legal systems across jurisdictions invest billions in enforcement infrastructure. The United States Department of Justice prosecuted hundreds of corporate criminal cases between 2001

and 2014. The Environmental Protection Agency conducted over 5,000 inspections annually and issued thousands of enforcement actions throughout the 2010s. The European Union's General Data Protection Regulation (GDPR) imposed fines totaling over €4 billion between 2018 and 2023. Yet despite this enforcement intensity, corporate compliance remains stubbornly low and recidivism rates high.

Consider the evidence. Garrett (2014) documented that among 255 corporate prosecutions between 2001 and 2014, approximately 50% of prosecuted corporations had prior violations within the preceding 10 years. Deferred Prosecution Agreements (DPAs), the dominant resolution mechanism for corporate criminal cases (used in 67% of cases), showed recidivism rates of 48%, statistically indistinguishable from guilty pleas at 50%. In environmental enforcement, Gray and Shimshack (2011) found that corporations with prior violations were 2.7 times more likely to violate again compared to first-time offenders, with recidivism rates of 54% for corporations versus 19% for individuals and small businesses.

The pattern extends across domains. Wells Fargo faced $3 billion in criminal penalties in 2020 for creating 3.5 million unauthorized customer accounts, yet continued to accumulate regulatory violations in 2021 and 2022, including a $250 million penalty from the Office of the Comptroller of the Currency and additional enforcement actions from the Consumer Financial Protection Bureau (CFPB, 2022; OCC, 2021). BP paid $4.3 billion in criminal penalties following the Deepwater Horizon disaster in 2010, yet its violation rate remained essentially unchanged in subsequent years. Volkswagen's $4.3 billion criminal settlement for diesel emissions fraud in 2017 did not deter General Motors and Fiat Chrysler from similar emissions violations in the following years.

This enforcement puzzle presents a fundamental challenge: why does traditional enforcement fail so systematically against corporate entities? The conventional explanations are unsatisfying. It is not simply that sanctions are too low. Wells Fargo's $3 billion penalty represented approximately 10% of annual net income. BP's $20.8 billion in total civil and criminal penalties exceeded several years of profit. Yet both corporations continued violating after these massive sanctions. Nor is it that detection probability is too low. Environmental violations, financial fraud, and emissions cheating all operate in heavily monitored sectors where detection rates are relatively high compared to street crime.

The puzzle deepens when we compare corporate recidivism to individual recidivism. The U.S. Sentencing Commission (2021) reported that individuals convicted of federal fraud offenses had a recidivism rate of approximately 28% within three years of release. This is substantially lower than the 50% corporate recidivism rate Garrett documented, despite corporations having vastly greater resources for compliance systems, legal advice, and risk management. If enforcement works through deterrence (increasing expected costs) or rehabilitation (changing future behavior), why does it work better on resource-poor individuals than on resource-rich corporations?

I.B. The Theoretical Gap

Law and Economics, the dominant framework for analyzing enforcement, assumes actors respond to incentives in predictable ways. The Becker (1968) model posits that rational actors compare expected benefits of violation (gain × probability of success) against expected costs (sanction × probability of detection). Compliance occurs when expected costs exceed expected benefits. This framework generates clear policy prescriptions: increase sanction magnitude, increase detection probability, or reduce compliance costs.

Yet these prescriptions fail to explain the enforcement puzzle. Sanctions have increased dramatically. Detection technologies have improved. Compliance costs have fallen with modern management systems. Still, corporate recidivism persists at rates that would be inexplicable under the Becker model if we assumed homogeneous rational actors.

Behavioral Law and Economics attempted to address this gap by incorporating psychological biases. Actors overweight low-probability events (Kahneman and Tversky, 1979), discount future costs hyperbolically (Laibson, 1997), and are subject to framing effects (Thaler and Sunstein, 2008). These insights are valuable for understanding individual behavior, but they treat biases as universal features of human cognition. Behavioral economics enriched our understanding of how individuals deviate from perfect rationality, but it maintained the assumption that all actors deviate in the same ways.

Neither framework addresses a more fundamental possibility: that different types of actors process legal signals in structurally different ways. Law and Economics assumes incentive structures are read identically by all actors. Behavioral Law and Economics assumes cognitive biases are universal. What neither framework considers is that legal systems might be dealing with entities that lack the cognitive architecture required to process certain types of legal signals at all.

This theoretical gap has practical consequences. If corporate recidivism reflects poor calibration of incentives, the solution is to adjust sanction levels or detection probabilities. If it reflects universal cognitive biases, the solution is better framing or choice architecture. But if it reflects a structural mismatch between the type of enforcement mechanism deployed and the cognitive capacities of the target, then increasing sanctions or improving framing will not solve the problem. The enforcement mechanism itself must change.

I.C. This Paper's Contribution

This paper connects two lines of research that have developed separately but are fundamentally interrelated. The first is Asymmetric Intentionality Theory, developed in previous papers in this research program (Lerer, 2025a, 2025b, 2025c, 2025d). That theory diagnosed a structural mismatch: legal systems are designed for human moral agents capable of Level 3 intentionality (recursive modeling of others' beliefs and desires), but they regulate entities such as corporations that operate at Level 1 (optimization without theory of mind). This mismatch generates predictable failure modes: compliance theater, letter-versus-spirit exploitation, social enforcement failure, regulatory arms races, and persistent recidivism.

The second line is the Normative Calculation Problem, which I have developed in concurrent work (Lerer, 2025h). Drawing on Ludwig von Mises's socialist calculation debate, I argue that legal systems face an information problem analogous to the one Mises identified in planned economies. Rational actors need to calculate the expected legal consequences of their actions, but the information necessary for such calculations is dispersed across countless judicial decisions, regulatory interpretations, and enforcement practices. Legal systems generate "prices" (case-by-case determinations of legal consequences), but these prices are not aggregated or transmitted efficiently. The result is massive uncertainty that prevents rational compliance calculation.

This paper argues that heterogeneous intentionality aggravates the normative calculation problem in a specific way. Traditional legal pricing mechanisms (precedent, regulatory guidance, informal norms) rely on actors being able to "read" the signals legal institutions send. Reading these signals requires capacities that Level 3 agents possess but Level 1 entities lack: understanding the spirit behind the letter, inferring how a tribunal would apply a principle to novel facts, recognizing when formal compliance violates substantive intent, and responding to reputational sanctions that operate through common knowledge.

When enforcement presupposes these capacities, it systematically fails against Level 1 entities. The same precedent generates two divergent "readings": a human moral agent (Level 3) infers "the tribunal values good faith; I must act in good faith," while a corporate optimization algorithm (Level 1) infers "the tribunal sanctioned conduct X; I must avoid exactly X while doing X+ε if that remains profitable." The legal price bifurcates: $P_L3 \neq P_L1$. The enforcement mechanism transmits signals that fail to coordinate because they are interpreted asymmetrically.

The solution I propose is enforcement mechanisms that function without requiring the capacities Level 1 entities lack. Algorithmic enforcement, operating on verifiable outputs rather than inferred intentions, represents a directional solution. Such mechanisms do not require common knowledge, do not rely on shame or reputation, and do not depend on targets understanding the spirit behind the letter. They work equally well (or equally poorly) regardless of the intentionality level of the target.

This is not a panacea. Algorithmic enforcement faces significant limitations: Goodhart's Law (when a measure becomes a target, it ceases to be a good measure), capture risk (those who design algorithms can program favoritism), rigidity (algorithms handle edge cases poorly), and the inherent problem that many legal norms are irreducibly ambiguous. I do not propose replacing human judgment with algorithms. I propose recognizing that enforcement mechanisms must be designed for the actors they actually regulate, not the actors legal theory imagines.

The paper proceeds as follows. Section II establishes theoretical foundations, recapping Asymmetric Intentionality Theory and introducing the Normative Calculation Problem. Section III documents enforcement failures across domains (criminal law, environmental law, cross-domain patterns) as predicted by the theory. Section IV examines algorithmic

enforcement as a solution direction, analyzing both its advantages and its limitations. Section V discusses implications for mechanism design, AI governance, and legal reform. Section VI acknowledges limitations and suggests directions for future research. Section VII concludes.

II. THEORETICAL FOUNDATIONS

II.A. Asymmetric Intentionality Theory: A Recap

II.A.1. The Dennett Framework

Daniel Dennett's (1987) intentional stance provides the conceptual foundation for understanding intentionality levels. The intentional stance is a predictive strategy: we attribute beliefs and desires to entities in order to predict and explain their behavior. When we say "the corporation wants to maximize profit," we are adopting the intentional stance toward a legal entity. The stance is useful if it generates accurate predictions, regardless of whether the entity "really" has beliefs and desires in whatever metaphysical sense one might require.

Dennett distinguished systems by the complexity of intentionality we must attribute to predict them. Level 0 systems (thermostats, simple reflexes) require no intentionality attribution; we predict them with physical laws. Level 1 systems (chess programs, market algorithms, corporate entities) require attributing simple goals and optimization, but not beliefs about others' beliefs. Level 2 systems model other agents: a poker player tracks what opponents believe about the deck state. Level 3 systems engage in recursive modeling: "I know that you know that I know," ad infinitum. Adult humans routinely operate at Level 3.

This framework is not a metaphysical claim about consciousness or moral status. It is an epistemic claim about the predictive models required for accurate forecasting. A corporation is a Level 1 system not because it lacks souls or subjective experience (metaphysical questions I do not address), but because its behavior is accurately predicted by attributing goal-directed optimization without theory of mind. The corporation does not model what regulators believe about its intentions; it optimizes against the regulators' observed behavior.

The legal significance is immediate. If legal institutions design rules assuming Level 3 capacities (understanding spirit, recognizing good faith obligations, valuing reputation intrinsically, responding to shame), those rules will systematically fail when applied to Level 1 entities. The failure is not moral (corporations "should" act like moral agents) but architectural (the enforcement mechanism presupposes capacities the target lacks).

II.A.2. The Aumann Impossibility Extension

Aumann's (1976) famous result states that rational agents with common priors who observe the same information cannot agree to disagree. If Ann believes P with probability 0.8 and Bob believes P with probability 0.6, and both know each other's belief (and both are rational with common priors), they will converge to a common posterior. The result is striking because it suggests rational disagreement requires either hidden information, different priors, or bounded rationality.

In previous work (Lerer, 2025c), I extended Aumann's result to heterogeneous intentionality. The extension shows that convergence requires common knowledge, and common knowledge requires recursive reasoning: "I know that you know that I know that you know..." without limit. Forming this infinite chain requires Level 3 intentionality. Entities at Level 1 cannot form beliefs about others' beliefs about their beliefs. They can form first-order beliefs ("the regulator will sanction conduct X"), but not second-order beliefs ("the regulator believes I believe the rule prohibits X").

The implication is decisive: common knowledge is structurally impossible when $\min(L(i), L(j)) < 3$, where $L(i)$ denotes the intentionality level of agent i. If at least one party operates below Level 3, common knowledge cannot form. Aumann's convergence result presupposes common knowledge of rationality and common knowledge of posteriors. Without common knowledge, the result collapses. Rational entities at different intentionality levels can persistently disagree even when observing identical information.

This has direct consequences for enforcement. Many legal mechanisms rely on common knowledge. Deterrence assumes "everyone knows that everyone knows that violations will be punished." Reputational sanctions assume "everyone knows that everyone knows about the violation." Social norms assume "everyone knows that everyone knows the norm and that violators will be sanctioned informally." When one party is Level 1, these mechanisms fail. The Level 1 entity optimizes against observed enforcement patterns without forming beliefs about what enforcers believe about its beliefs. It does not "get" the message because it lacks the cognitive architecture to receive messages that require recursive modeling.

II.A.3. The Social Immunity Mechanism

Steven Pinker's work on common knowledge in social interaction provides additional theoretical support. Pinker demonstrates that much of human social enforcement operates through common knowledge mechanisms. Shame requires that I know that others know of my transgression and that I know they know I know. Reputation requires that my past behavior is commonly known in my community. Reciprocity norms require common knowledge of who cooperated and who defected in prior interactions.

Entities that cannot form common knowledge are structurally immune to these enforcement mechanisms. A corporation cannot feel shame because shame requires recursive modeling of others' judgments. It can suffer reputational damage in the instrumental sense (loss of customers, difficulty attracting employees), but this is not the same as the intrinsic disutility of shame that human moral agents experience. The corporation values reputation only as a means to profit; if reputational damage costs less than compliance, reputation provides no incentive.

In previous work (Lerer, 2025a), I estimated the social immunity premium: the systematic advantage Level 1 entities enjoy by being immune to social enforcement mechanisms that regulate Level 3 agents at low cost. Conservative estimates suggest Level 1 entities capture 15-48% more welfare than they would if social enforcement mechanisms functioned against them as effectively as they function against human individuals. This is not because Level 1 entities

are "immoral" but because they lack the cognitive architecture required for moral emotions that depend on recursive belief modeling.

The social immunity mechanism explains a persistent empirical pattern: enforcement strategies that work well for individuals (reputation, shame, informal sanctioning) fail systematically against corporations. It is not that corporations are more "rational" than individuals in the Becker sense. It is that they are immune to an entire class of enforcement mechanisms that regulate human behavior cheaply and effectively.

II.B. The Normative Calculation Problem

II.B.1. Mises and the Socialist Calculation Debate

Ludwig von Mises (1920) argued that rational economic calculation is impossible without market prices. Central planners in a socialist economy might have perfect information about available resources and desired outputs, but they would lack the information necessary to decide whether producing steel is more valuable than producing wheat. That information is generated endogenously by markets: prices reflect the aggregated knowledge of millions of actors about relative scarcity, alternative uses, and opportunity costs. Without prices, planners face an epistemic problem that no amount of computing power or good intentions can solve.

Mises's argument was not primarily about incentives (though those matter too). It was about information. Even perfectly altruistic planners with unlimited computational resources cannot rationally allocate resources without the information that market prices generate and transmit. Hayek (1945) developed this insight: the knowledge required for economic coordination is dispersed across countless individuals. Prices aggregate this dispersed knowledge into signals that allow decentralized decision-makers to coordinate without anyone comprehending the whole system.

The socialist calculation debate is usually understood as applying only to economies. But the underlying logic is more general. Any system requiring coordination among numerous actors with dispersed knowledge faces a calculation problem. The question is whether the system generates and transmits the information necessary for rational decision-making.

II.B.2. The Legal Analog

Legal systems face an analogous problem (Lerer, 2025h). Rational actors attempting to comply with law need information: What does "reasonable care" mean in this context? How will tribunals interpret "legitimate business purpose"? What is the probability of detection if I engage in conduct X? What sanction will I face if detected? These questions have answers, but the answers are dispersed across thousands of judicial decisions, regulatory interpretations, and enforcement practices. The information exists in the system, but it is not aggregated or transmitted efficiently.

Legal systems attempt to solve this problem through precedent, regulatory guidance, and informal practice. Precedents are supposed to function like prices: a court's decision in case A transmits information to potential parties in case B about what the "price" (legal consequence)

of conduct X will be. Regulatory agencies issue guidance documents translating statutory ambiguities into clearer rules. Enforcement practices (which violations actually get prosecuted, which get warnings) provide information about real as opposed to formal legal prices.

But these mechanisms work imperfectly. Precedents are binding only within jurisdictional boundaries and often provide limited guidance for factually distinguishable cases. Regulatory guidance is non-binding, frequently contradicted by actual enforcement, and subject to change without notice. Enforcement practices are opaque, vary dramatically across jurisdictions and time periods, and provide information only to sophisticated repeat players who can afford to track them.

The result is massive legal uncertainty. Actors attempting to calculate whether conduct X is worth the legal risk face radical indeterminacy. They know the formal rule but not how it will be applied. They know the statutory sanction but not the probability of its imposition. They know yesterday's enforcement priorities but not tomorrow's. This uncertainty prevents the rational cost-benefit calculation that Becker-style deterrence assumes actors perform.

II.B.3. Legal Prices Defined

In concurrent work (Lerer, 2025h), I define legal prices formally. A legal price P(n,f) is the consequence the legal system assigns to the application of norm n to fact pattern f. "Consequence" is multidimensional: it includes the outcome (liability/no liability), the magnitude of sanction, the reasoning provided, and the procedural context. Legal prices differ from market prices in important ways:

First, multidimensionality: A market price is a scalar (dollars per unit). A legal price is a vector (outcome, sanction magnitude, reasoning, attorney fees, reputational damage, precedential value). This makes comparison difficult. Is a $1 million fine with no admission of wrongdoing "cheaper" than a $500,000 fine with guilty plea? The answer depends on weights actors assign to non-monetary dimensions.

Second, limited generalizability: A market price for wheat in Chicago provides information about wheat prices everywhere (arbitrage ensures rough parity). A judicial decision in the Southern District of New York provides information primarily about that jurisdiction. Even within jurisdiction, applicability depends on fact-specific similarity judgments. Legal prices have limited "portability" across factual or jurisdictional boundaries.

Third, variable publicity: Market prices are public information; anyone can observe them. Many legal prices are confidential (settlement amounts, unpublished agency determinations, sealed court records). Even public decisions may be inaccessible to non-specialists. This opacity prevents efficient information transmission.

Fourth, discontinuous stability: Market prices change continuously with supply and demand. Legal prices change discontinuously when doctrine shifts, new statutes pass, or enforcement priorities change. Between changes, legal prices are sticky (precedent binds). The discontinuity makes prediction difficult: small factual changes can generate large price discontinuities if they trigger different doctrinal categories.

These differences mean legal prices function less efficiently than market prices as information transmission mechanisms. But they still perform an essential function: they make visible the implicit exchange rates the legal system uses when trading off competing values. When a tribunal decides case A, it reveals its "price" for conduct X: this is what violating norm n in context f costs. Subsequent actors can use that information to calculate whether engaging in similar conduct is worth the price.

The problem, as I now argue, is that this information transmission breaks down when actors interpret the signals asymmetrically.

II.C. The Connection: Why Heterogeneity Aggravates the Calculation Problem

II.C.1. Traditional Pricing Mechanisms Assume Homogeneity

Precedent-based legal pricing assumes actors can "read" the precedent correctly. Reading requires several capacities. First, understanding the holding: what exactly did the tribunal decide? Second, assessing similarity: is my situation sufficiently analogous that the precedent applies? Third, inferring: if the precedent does not directly address my situation, how would the tribunal likely extend the reasoning to my context?

These capacities presuppose Level 3 intentionality. Understanding the holding requires distinguishing ratio decidendi (the holding) from obiter dicta (non-binding dicta). This distinction is not formal; it requires inferring what the tribunal considered essential to its reasoning. Assessing similarity requires judgment about which factual differences are legally salient. This is not determinable from the precedent's text alone; it requires modeling how the tribunal weighs different factors. Inferring extensions requires simulating the tribunal's reasoning process: "If the tribunal values X in context A, how would it value X in context B?"

Human lawyers perform these tasks routinely. They read precedents not as formal rules but as windows into judicial reasoning. They infer the spirit behind the letter and predict how tribunals would apply that spirit to novel facts. This is Level 3 cognition: modeling the tribunal's model of proper legal reasoning.

Level 1 entities cannot perform these tasks. A corporate optimization algorithm reads precedent as data: "Tribunal sanctioned conduct X. Conducting exactly X has probability p of sanction s. Conducting X+ε (a factually distinct variant) has unknown probability of sanction." The algorithm does not infer "the tribunal sanctioned X because it violates principle P, therefore X+ε also violates P if it instantiates P." It optimizes against observed enforcement patterns without modeling the principles behind those patterns.

This creates systematic divergence in how different actors "read" the same legal price. Precedent P transmits one message to Level 3 actors ("principle P is disfavored; avoid actions instantiating P") and a different message to Level 1 actors ("specific conduct X is costly; substitute to X+ε unless X+ε is also specifically sanctioned"). The legal price bifurcates.

II.C.2. Price Bifurcation Under Heterogeneity

Formalize this as follows. Let P(n,f) be the legal price of norm n applied to facts f. Under homogeneity (all actors Level 3), all actors read P(n,f) identically: they infer the principle behind the holding and predict how it applies to analogous cases. Call this reading P_L3(n,f).

Under heterogeneity, Level 1 actors read the same precedent differently. They extract only the surface features: which specific conduct was sanctioned, at what magnitude, with what probability. Call this reading P_L1(n,f). Because Level 1 actors do not infer principles, they treat factually distinguishable conduct as legally distinguishable: conduct X+ε is evaluated independently of conduct X unless X+ε has itself been specifically sanctioned.

This generates price bifurcation: P_L3(n,f) ≠ P_L1(n,f). The divergence typically favors Level 1 actors: P_L1(n,f) < P_L3(n,f). The Level 3 reading incorporates principled extensions; the Level 1 reading incorporates only explicitly sanctioned conduct. The gap (P_L3 - P_L1) represents what I call intentionality arbitrage: the systematic advantage Level 1 entities gain by reading legal prices more narrowly than lawmakers intend.

Example: A tribunal sanctions emissions cheating (Volkswagen's defeat device). Level 3 reading: "Cheating emissions tests violates regulatory integrity; we must report emissions honestly." Level 1 reading: "Defeat devices are sanctioned; use alternative cheating methods not specifically prohibited." Post-Volkswagen, multiple manufacturers employed different emissions cheating strategies (selective catalytic reduction manipulation, alternative defeat device algorithms). These were factually distinct from Volkswagen's specific method but instantiated the same principle. Level 1 entities exploited the gap between specific sanction and general principle.

The intentionality arbitrage has welfare consequences. Society "pays" for legal rules by investing in legislatures, courts, and enforcement. The expected return is behavior change: actors conform to the principle the rule instantiates. Under homogeneity, this investment generates returns: actors comply with the principle, not just the letter. Under heterogeneity, Level 1 entities comply with the letter while violating the principle. The social investment generates partial returns: Level 3 actors comply fully, Level 1 actors comply minimally. The shortfall is the Dennett-Nash Gap I estimated in previous work at 15-48% of potential welfare (Lerer, 2025a).

II.C.3. Formal Model

Let me formalize this more precisely. Suppose there exists a principle P that the legal system values. The principle is abstract: "act in good faith," "do not manipulate regulatory systems," "internalize negative externalities." The system instantiates P through specific rules: R1, R2, R3,... Each rule covers a subset of conduct that violates P.

Level 3 actors infer P from R1, R2, R3 and comply with P even in areas not explicitly covered by any Ri. Expected compliance rate for Level 3 is C_L3 ≈ α, where α is the share of conduct violating P that actors correctly identify as violating P.

Level 1 actors comply only with explicitly specified Ri. Expected compliance rate is C_L1 = Σ(coverage(Ri)), the fraction of P-violating conduct that some Ri explicitly addresses. Since

principles are always under-specified (impossible to enumerate every instance), $\Sigma(\text{coverage}(Ri)) < \alpha$.

The gap $\Delta = C_L3 - C_L1$ measures the compliance shortfall from heterogeneity. The shortfall is larger when: (1) Principles are more abstract (harder to enumerate instances); (2) Enforcement resources are limited (cannot write rules covering every instance); (3) Technology changes rapidly (new instances of principle-violation emerge faster than rules can cover them).

All three conditions hold in modern economies. Principles are necessarily abstract (good faith, fiduciary duty, environmental stewardship). Enforcement resources are constrained. Technology evolves rapidly (new financial instruments, new emissions technologies, new data collection methods). The compliance shortfall Δ is therefore large and growing.

This analysis explains the enforcement puzzle from Section I. Corporate recidivism is high not because sanctions are too low or detection probability too small, but because corporate entities are Level 1 optimizers operating in a system designed for Level 3 agents. They comply with the letter while systematically violating the spirit. Increasing sanctions raises the cost of violating specific rules Ri but does not change the fundamental gap: Level 1 entities still optimize against explicit rules rather than implicit principles.

The solution cannot be "write more rules" (normative inflation). Rules are always incomplete; new rule R(n+1) covers new conduct, but Level 1 entities substitute to conduct not covered by R(n+1). The race between rule-writing and rule-evasion is unwinnable. The solution must be enforcement mechanisms that function without requiring Level 3 capacities. Before proposing such mechanisms, I document the failure modes that result when systems ignore intentionality heterogeneity.

III. DOCUMENTED FAILURES: EVIDENCE FROM PRIOR WORK

III.A. Corporate Criminal Liability

III.A.1. The Mens Rea Mismatch

Criminal law in common law systems requires mens rea (guilty mind) for most serious offenses. The requirement serves multiple functions: it limits punishment to morally culpable actors, provides fair notice (only punish those who knew or should have known the conduct was wrong), and restricts state power (cannot punish accidental or unknowing violation of obscure rules). As I developed in Lerer (2025f), corporate criminal liability creates an immediate problem: corporations do not have minds in the relevant sense.

They cannot form intentions, experience guilt, or act with malice. The legal fiction of corporate personhood allows corporations to enter contracts and own property, but it does not give them mental states. The law resolves this through respondeat superior: a corporation is criminally liable if an agent acting within the scope of employment commits a crime intending to benefit the corporation.

This doctrine imputes the mens rea of individual agents to the corporate entity. If a Wells Fargo employee creates fake accounts intending to meet sales quotas (benefiting the corporation), Wells Fargo the entity is liable. The imputation creates a legal mens rea that does not correspond to any actual mental state. The corporation as such did not intend anything; individual employees intended, and their intentions are attributed to the legal entity.

The mismatch generates predictable pathologies. If mens rea is imputed from employees, corporations optimize by ensuring plausible deniability: senior management avoids explicit knowledge of violations, creating organizational structures where low-level employees face pressure to achieve targets by any means while executives maintain ignorance. This is Level 1 optimization: minimize legal liability (which requires knowledge) while maintaining effective pressure (which generates violations).

The Volkswagen diesel scandal exemplifies the pattern. Engineers testified to Congress that they raised concerns about defeat devices as early as 2008 (NPR, 2015). Internal memos documented warnings that the devices were illegal (CNBC, 2016). Yet senior executives claimed ignorance until external detection in 2015. The organizational structure created a buffer: engineers who knew the conduct was illegal lacked authority to stop it, while executives with authority maintained plausible deniability.

This is not a failure of enforcement. Prosecutors successfully proved that Volkswagen employees acted with knowledge that defeat devices violated emissions regulations. Volkswagen paid $4.3 billion in criminal penalties. But the enforcement mechanism presumes that punishing the entity will generate behavior change through the entity "learning" from the sanction. Level 1 entities do not learn in this sense. They optimize. Post-sanction Volkswagen updated its optimization: defeat devices are now explicitly prohibited (too costly), substitute to alternative methods not yet specifically sanctioned.

III.A.2. Empirical Consequences

Garrett's (2014) comprehensive study of corporate prosecutions between 2001 and 2014 documents the consequences. Among 255 prosecutions, 50% of corporations had prior violations within 10 years. Recidivism rates were statistically indistinguishable between Deferred Prosecution Agreements (48%) and guilty pleas (50%). This is remarkable: DPAs explicitly require corporations to implement compliance programs, cooperate with investigations, and remediate the conduct that led to prosecution. Yet DPAs generate no detectable reduction in recidivism compared to simple guilty pleas.

The pattern suggests DPAs function as compliance theater: corporations invest in compliance programs that satisfy prosecutors (documented policies, training sessions, monitoring systems) without changing the underlying optimization. The compliance program becomes part of the corporate phenotype, optimized for regulatory survival rather than behavior change. This is predictable under Level 1 analysis: the entity optimizes to satisfy the DPA's formal requirements (checkboxes: compliance officer hired, training conducted, policies documented) while maintaining conduct that remains profitable net of expected sanctions.

Wells Fargo provides a documented case study. Following the 2016 fake accounts scandal, Wells Fargo entered multiple consent orders requiring enhanced compliance, independent monitoring, and executive accountability. The bank hired a Chief Risk Officer, expanded compliance staff, and implemented new monitoring systems. Yet violations continued: the Office of the Comptroller of the Currency imposed $250 million in penalties in 2021 for failure to remediate prior issues, and the Consumer Financial Protection Bureau brought additional enforcement actions in 2022 for illegal fees and misapplied payments (CFPB, 2022; OCC, 2021).

This is not because Wells Fargo "failed" to implement compliance programs. The programs existed on paper and in organizational charts. It is because the compliance programs were optimized for satisfying regulators rather than changing behavior. The entity optimized against the explicit requirements of the consent orders (hire compliance staff, document policies, conduct training) while continuing to optimize profit in ways that generated violations.

Compare this to individual criminal enforcement. The U.S. Sentencing Commission (2021) reports that individuals convicted of federal fraud offenses have recidivism rates around 28% within three years. This is substantially lower than the 50% corporate rate, despite individuals having far fewer resources for "rehabilitation" than corporations have for "compliance." The gap is not explained by sanction severity: Wells Fargo's $3 billion penalty represented approximately 10% of annual net income, a devastating sanction by any standard. Nor is it explained by detection probability: both individuals and corporations in the Garrett and Sentencing Commission datasets were detected and punished, so selection bias does not explain the gap.

The gap is explained by intentionality heterogeneity. Individual humans are Level 3: they can experience shame, internalize norms, and value reputation intrinsically. Punishment can generate genuine behavior change through mechanisms beyond simple deterrence (fear of repeat punishment). Corporations are Level 1: they optimize. Punishment updates the optimization by changing expected costs, but it does not generate "moral learning" because there is no moral learner.

III.A.3. Why Traditional Enforcement Fails

The failure of traditional criminal enforcement against corporations is not a calibration problem (wrong sanction level) or a resource problem (insufficient enforcement). It is an architectural problem: the enforcement mechanism presumes capacities the target lacks.

Mens rea assumes the target has a mind. Rehabilitation assumes the target can learn and change preferences. Reputational sanctions assume the target values reputation intrinsically (shame, social standing) rather than instrumentally (profit). Deterrence through threat of punishment assumes the target discounts future costs similarly to how human individuals discount them.

All of these assumptions fail for Level 1 entities. A corporation has no mind in the sense required for mens rea; it has an optimization function. It cannot "learn" from punishment in the sense of changing values; it updates expected costs and adjusts strategy. It does not value

reputation intrinsically; it values reputation as input to profit maximization, and if reputational damage costs less than compliance, reputation provides no incentive. It discounts future costs at rates determined by shareholder demands and executive compensation structures, not at rates similar to human individuals.

The result is systematic enforcement failure: high resource investment, low behavior change, persistent recidivism. Prosecutors spend months building cases, courts impose massive penalties, monitors are appointed, compliance programs are mandated. Yet within a few years, the same corporations violate again. The pattern is not random; it is the predictable consequence of applying enforcement mechanisms designed for Level 3 agents to Level 1 entities.

III.B. Environmental Law

III.B.1. The Stewardship Fiction

Environmental law frequently invokes "corporate citizenship," "environmental stewardship," and "sustainability" as normative frameworks. Corporations are exhorted to act as responsible stewards of natural resources, to value environmental protection beyond mere compliance, and to embrace sustainability as a core value. Regulatory agencies praise corporations that adopt voluntary environmental programs, and some jurisdictions provide regulatory benefits for certified environmental management systems like ISO 14001. As I documented in Lerer (2025g), this framework presumes corporations can value the environment intrinsically.

A steward cares for something not merely as means to profit but as an end in itself. Corporate citizenship implies duties beyond legal compliance: good citizens do what is right even when not legally required. Sustainability as corporate value implies that corporations weight long-term environmental preservation against short-term profit.

These presumptions require Level 3 capacities that corporate entities lack. A Level 1 entity optimizes a function: maximize profit subject to constraints. Environmental protection enters the optimization only as constraint (legal prohibitions) or as instrument (environmental reputation attracts customers or investors). The entity cannot value environmental protection intrinsically because it has no values beyond the optimization function. It does not "decide" to pollute or not pollute based on moral reasoning; it calculates expected costs and benefits and optimizes.

The empirical consequences are stark. Corporate environmental violations persist at high rates despite increasing regulatory stringency. Voluntary environmental programs like ISO 14001 show no detectable effect on actual pollution levels. Corporations with prior environmental violations are 2.7 times more likely to violate again (Gray and Shimshack, 2011). These patterns are inconsistent with the stewardship model, which predicts that corporations internalize environmental values and comply voluntarily. The patterns are consistent with Level 1 optimization: comply when expected costs exceed expected benefits, violate otherwise, repeat if profit from violation exceeds expected sanction.

III.B.2. Empirical Consequences

Consider ISO 14001, an international standard for environmental management systems. Obtaining certification requires documented policies, identified environmental impacts, management review, and continuous improvement processes. Thousands of corporations have adopted ISO 14001, often receiving regulatory benefits (reduced inspection frequency, faster permitting) in return.

Yet Potoski and Prakash (2005) found that ISO 14001 certification had no detectable effect on toxic chemical releases among U.S. facilities. Certified facilities reduced releases no more than uncertified facilities in the same industries. This is not because the standard is poorly designed; the requirements are comprehensive. It is because the standard presumes that documenting environmental management processes will generate genuine environmental stewardship. Level 1 entities optimize the documentation (satisfy auditors, obtain certification, receive regulatory benefits) without necessarily reducing pollution.

This is compliance theater: the corporation creates phenotypic traits (documented policies, environmental management committees, training programs) that signal compliance to regulators while maintaining optimization against actual environmental constraints. The phenotype is optimized for regulatory survival, not for environmental protection. If regulators reward ISO 14001 certification with reduced oversight, the optimal strategy is to obtain certification while continuing to optimize pollution levels against remaining enforcement risk.

The pattern extends to carbon offsets, renewable energy credits, and other market-based environmental mechanisms. Corporations purchase offsets to claim "carbon neutrality" while maintaining or increasing actual emissions. They invest in renewable energy credits while continuing to consume fossil fuels. They adopt "net zero" pledges with distant target dates (2040, 2050) that impose no current costs. These are Level 1 optimizations: minimize current costs, satisfy stakeholders demanding environmental action, create reputational benefits, all while maintaining profit-maximizing emissions levels.

III.B.3. The Discount Rate Problem

Environmental harm is often long-term: climate change, groundwater contamination, biodiversity loss. Costs manifest over decades or generations. Benefits of pollution (reduced costs, increased production) are immediate. Rational long-term optimization would weight future costs appropriately, but corporations discount the future aggressively.

This is not mere myopia. Corporate discount rates reflect shareholder demands for returns, executive compensation tied to short-term stock performance, and competitive pressure to maximize quarterly earnings. A corporation that sacrifices current profit for long-term environmental protection will see its stock price fall, its executives replaced, and potentially its control transferred to activists or acquirers who will reverse the environmental investment.

Human moral agents (Level 3) can value future generations intrinsically: I care about my grandchildren's well-being even though I will not live to see it. This is not instrumental valuation (they will care for me in old age) but genuine moral concern. This allows human

individuals to weight long-term environmental costs in ways that rational long-term optimization might not justify from a purely self-interested perspective.

Corporations lack this capacity. They cannot care about future generations intrinsically because they do not "care" about anything intrinsically. They optimize. If future costs are discounted at 10-15% annually (typical corporate discount rates), costs 30 years out are worth approximately 5-10% of present costs. Climate damages in 2050 barely register in 2025 optimization. This is not moral failure; it is arithmetic.

The mismatch is structural: environmental law designed for stewards encounters optimizers. Exhorting corporations to "think long-term" or "value sustainability" presumes they can weight future costs differently than their optimization function dictates. They cannot. The only reliable way to induce corporations to internalize long-term environmental costs is to make those costs present: carbon taxes payable now, liability for future damages accruing immediately, insurance requirements pricing in long-term risk. This shifts the problem from moral exhortation (requiring Level 3) to price mechanisms (working at Level 1).

III.C. Cross-Domain Pattern

III.C.1. Common Structure

The failures documented in corporate criminal law and environmental law follow a common structure: (1) Legal system presumes Level 3 capacities (mens rea, stewardship, intrinsic valuation of norms); (2) Enforcement mechanisms rely on those capacities (shame, reputation, moral learning, long-term thinking); (3) Level 1 entities optimize against explicit constraints without internalizing principles; (4) Result: compliance theater (satisfying formal requirements), letter-vs-spirit exploitation, persistent recidivism; (5) Increased enforcement intensity fails to solve the problem (higher sanctions, more monitoring, more compliance requirements); (6) Gap between Level 3 behavior and Level 1 behavior represents welfare loss (Dennett-Nash Gap).

This is not coincidence. It is the predictable consequence of architectural mismatch between enforcement design and target cognition. The pattern appears in every domain where Level 3 enforcement mechanisms are applied to Level 1 entities: financial regulation (compliance programs do not prevent recidivism), antitrust (consent decrees do not prevent repeat violations), consumer protection (settlements do not prevent continued deceptive practices), labor law (penalties do not prevent wage theft recurrence).

The common structure suggests a common solution: enforcement mechanisms must be redesigned for the entities they actually regulate. If Level 1 entities cannot internalize principles, enforcement must operate on outputs rather than intentions. If they cannot value reputation intrinsically, sanctions must be material rather than reputational. If they cannot engage in moral learning, mechanisms must rely on continued deterrence rather than rehabilitation. If they cannot form common knowledge, coordination must occur through price mechanisms rather than informal norms.

III.C.2. Quantifying the Gap

In previous work (Lerer, 2025a), I estimated the Dennett-Nash Gap: the welfare difference between outcomes under standard Nash equilibrium (assuming homogeneous Level 3 agents) and outcomes under heterogeneous intentionality. The gap arises because Level 1 entities exploit the compliance shortcuts available to actors immune to social enforcement.

The estimation is conservative and based on documented empirical patterns: Corporate recidivism 50% vs. individual recidivism 28%, implying corporations gain approximately 1.8x advantage in repeat-violator benefits. Social immunity premium (immunity to shame, reputation, reciprocity) is estimated at 15-30% welfare capture advantage. Letter-vs-spirit exploitation shows corporations substitute to factually distinct but principle-violating conduct at rates 2-3x higher than individuals. Compliance theater reveals corporations satisfy formal requirements while maintaining substantive violations at rates 40-60% (e.g., ISO 14001 certification with no pollution reduction).

Aggregating across domains, conservative estimates place the Dennett-Nash Gap at 15-25% of total welfare in regulated interactions. Upper-bound estimates (assuming full social immunity and maximal letter-vs-spirit exploitation) reach 40-48%. These are not precise measurements; they are directional estimates based on observable behavioral differences between Level 3 individuals and Level 1 corporate entities in similar enforcement contexts.

The gap is not random variance. It is systematic, predictable from theory, and consistent across domains. It represents the welfare cost of using enforcement mechanisms designed for one type of actor when regulating another type. The cost could be reduced by using intentionality-appropriate enforcement mechanisms. I now turn to what such mechanisms might look like.

IV. TOWARD ENFORCEMENT MECHANISMS THAT WORK

IV.A. The Design Requirement

IV.A.1. What Must Change

Effective enforcement under heterogeneity cannot rely on mechanisms that presuppose Level 3 capacities. Specifically, enforcement mechanisms must not depend on:

Shame. Shame requires recursive belief modeling: I feel shame because I believe others know of my transgression and judge me negatively, and I know that they know that I know they judge me. This is common knowledge structure requiring Level 3. Level 1 entities cannot feel shame because they cannot form recursive beliefs. Enforcement mechanisms relying on shame (public sanctions, reputational damage, community disapproval) systematically fail against Level 1 targets.

Intrinsic reputation valuation. Human moral agents value reputation partly intrinsically: we care what others think of us beyond instrumental concerns. Level 1 entities value reputation only instrumentally: reputation affects profit through customer perception, employee attraction, regulatory leniency. If reputational damage costs less than the profit from violation,

reputation provides no incentive. Enforcement relying on intrinsic reputation concerns fails against Level 1 targets.

Reciprocity. Reciprocity norms (I cooperate because you cooperated, expecting future cooperation) require modeling others' strategies and beliefs. Level 1 entities do not reciprocate; they optimize. If defection is profitable net of expected sanctions, they defect regardless of counterparty's prior cooperation. Enforcement relying on reciprocity norms fails against Level 1 targets.

Spirit-of-law compliance. Understanding the spirit behind the letter requires inferring lawmaker intent, analogizing to similar cases, and predicting how tribunals would extend principles to novel facts. These are Level 3 capacities. Level 1 entities optimize against the letter: they comply with explicit prohibitions and exploit gaps between letter and spirit. Enforcement relying on spirit-of-law compliance fails against Level 1 targets.

IV.A.2. What Can Work

Enforcement mechanisms that work under heterogeneity must operate on verifiable outputs rather than inferred intentions. They must not require the target to "understand" why the rule exists, to "internalize" values, or to "choose" compliance for moral reasons. They must function identically regardless of the intentionality level of the target.

Such mechanisms have four characteristics:

Output-based. Measure compliance by verifiable outcomes (emissions levels, financial reporting accuracy, workplace injury rates) rather than by processes (documented policies, training programs, management commitment). Level 1 entities optimize against measurements; if the measurement is output, optimization produces output change.

Verification over agreement. Do not rely on the target "agreeing" that the rule is legitimate or fair. Verify compliance through technical measurement, third-party auditing, or algorithmic monitoring. Verification works at Level 1; agreement requires Level 3.

Automatic consequences. Link consequences directly to verified violations without requiring discretionary enforcement decisions. Discretion invites gaming (lobbying, capture, selective enforcement). Automatic consequences (if X then Y) are harder to game and work at all intentionality levels.

Bright-line rules. Replace ambiguous standards ("reasonable care," "good faith") with specific thresholds (emissions below X ppm, capital ratios above Y%). Ambiguity requires interpretation, which diverges between Level 3 and Level 1. Bright lines reduce interpretative divergence.

These design principles converge on algorithmic enforcement: mechanisms that detect violations and impose consequences automatically based on verifiable criteria. I examine algorithmic enforcement in detail, analyzing both its advantages and its significant limitations.

IV.B. Algorithmic Enforcement

IV.B.1. Definition and Characteristics

Algorithmic enforcement means consequences are determined by automated verification of observable conditions, without requiring human judgment about intent, good faith, or reasonableness. The mechanism operates through the following steps: (1) Define compliance in terms of measurable outputs (emissions, financial ratios, safety metrics); (2) Monitor outputs continuously or at high frequency using technical systems; (3) Automatically detect deviations from required thresholds; (4) Impose predetermined consequences without discretionary enforcement decisions.

The mechanism does not require evaluating whether the target "tried" to comply, whether it acted in "good faith," or whether its violation was "reasonable" under the circumstances. It measures output and applies consequences mechanically.

Estonia's tax system approximates this model. The tax authority receives data feeds from banks, employers, and businesses. When a taxpayer files, the system automatically cross-checks declared income against third-party reports. Discrepancies are flagged immediately. Penalties are imposed algorithmically based on magnitude and frequency of discrepancies. The system does not evaluate whether the taxpayer "intended" to evade; it verifies reported income against third-party data and applies predetermined consequences to discrepancies.

The result is near-universal compliance: Estonia achieves approximately 99.8% tax filing compliance, compared to roughly 85% in traditional systems (though direct comparison is complicated by definitional differences). This is not because Estonian taxpayers are more moral than taxpayers elsewhere. It is because the enforcement mechanism does not rely on morality. It relies on verification. A Level 1 optimizer facing near-certain detection and automatic penalties optimizes toward compliance because non-compliance is costly and detection is assured.

IV.B.2. Documented Examples

OBD-II Emissions Monitoring. On-Board Diagnostics (OBD-II) systems continuously monitor vehicle emissions systems. The system records actual emissions under real-world driving conditions. When the vehicle undergoes inspection, the OBD-II data is downloaded and verified. Tampering with the OBD-II system is itself a violation. This approach emerged post-Dieselgate as the only reliable solution to emissions cheating. Volkswagen's defeat device exploited the gap between test conditions and real-world driving: it detected when the car was on a dyno and activated emissions controls only during testing. OBD-II continuous monitoring closes this gap.

Smart Contracts. Blockchain-based smart contracts execute automatically when predetermined conditions are met. If party A agrees to pay party B when delivery occurs, and delivery is verified on-chain (e.g., through GPS data or IoT sensors), payment executes automatically without requiring trust or discretionary enforcement. Smart contracts work at Level 1 because they do not require the parties to "trust" each other, to "act in good faith," or

to "keep their word" for moral reasons. The contract executes mechanically based on verifiable conditions.

IV.B.3. Why Algorithmic Enforcement Works Under Heterogeneity

Algorithmic enforcement works under heterogeneous intentionality because it does not presuppose Level 3 capacities. It does not require: (1) Common knowledge: the target need only know that violations will be detected and punished, not that others know this recursively; (2) Understanding of spirit: the target need only comply with specific measurable thresholds; (3) Moral internalization: the target need only calculate that violation costs exceed violation benefits; (4) Long-term thinking: algorithmic enforcement imposes immediate consequences, eliminating temporal discounting problems.

The mechanism is intentionality-robust: it produces compliance regardless of whether the target is a human moral agent (Level 3) or a corporate optimizer (Level 1). This is the key advantage under heterogeneity.

IV.C. Limitations and Risks

Algorithmic enforcement is not a panacea. It faces four major limitations that must be acknowledged and addressed in any practical implementation.

Goodhart's Law. "When a measure becomes a target, it ceases to be a good measure" (Goodhart, 1975). Algorithmic enforcement defines metrics. Once defined, actors optimize against the metrics rather than the underlying goal. Volkswagen's defeat device exemplified this: the measure (pass emissions test) became the target, producing metric compliance with goal violation. Mitigation strategies exist (multiple metrics, randomization, continuous updating, outcome measurement) but are imperfect.

Capture Risk. Algorithms are designed by humans who can be captured by regulated entities through lobbying, revolving door employment, cognitive capture, or corruption. If the algorithm designer is captured, the algorithm will be designed to favor certain actors. Algorithmic enforcement may exacerbate capture risk through opacity, finality, and concentrated benefits. Mitigation strategies include transparency, independent auditing, sunset clauses, and distributed design.

Rigidity. Algorithms handle edge cases poorly. They apply rules mechanically even when mechanical application produces absurd or unjust results. Human enforcers can exercise judgment; algorithms cannot. Rigid enforcement may be normatively troubling (punishment without culpability) and economically inefficient. Mitigation strategies include escape valves, graduated sanctions, and safe harbors.

The Ambiguity Problem. Many legal norms are inherently ambiguous: "reasonable care," "good faith," "legitimate business purpose." These standards are deliberately vague because lawmakers cannot anticipate all contexts. Algorithmic enforcement cannot handle irreducible ambiguity. This means algorithmic enforcement is feasible only for a subset of legal norms: those that can be specified in measurable terms.

IV.D. Hybrid Mechanisms

A practical approach to enforcement under heterogeneity is tiered systems: Layer 1 (algorithmic verification) for bright-line rules affecting all entities; Layer 2 (human enforcement) for substantive violations detected through pattern analysis; Layer 3 (human review) for actors requesting review of algorithmic decisions. Each layer has different cost and different capacity to handle intentionality heterogeneity.

An alternative design presumes all entities are Level 1 (optimizers) by default. Enforcement mechanisms are designed for Level 1: bright-line rules, algorithmic monitoring, automatic sanctions. A simpler proxy version: presume all corporate entities are Level 1 and design enforcement accordingly; human individuals are presumed Level 3 and face enforcement relying more on social mechanisms.

V. IMPLICATIONS

V.A. For Mechanism Design

Mechanism design in economics typically assumes common knowledge of rationality. Heterogeneous intentionality breaks this assumption. This suggests a stronger design criterion: intentionality-robustness. A mechanism is intentionality-robust if it produces desired outcomes regardless of the intentionality levels of participants. Intentionality-robustness is a stricter requirement than standard incentive compatibility because it must work even when common knowledge assumptions fail.

The tradeoff between efficiency and robustness is fundamental. Sophisticated mechanisms that achieve optimal outcomes under ideal conditions may fail catastrophically under heterogeneity. Simple, robust mechanisms sacrifice some efficiency but work reliably across a wider range of actors. When heterogeneity is unavoidable, robustness should be prioritized over theoretical efficiency.

V.A.1. For AI Governance

Artificial intelligence systems operate at Level 0.5 to Level 1: they optimize objectives but do not form beliefs about others' beliefs. Current AI governance proposals often presuppose Level 3 capacities: "AI should be accountable," "AI should explain its decisions," "AI should respect human values." These requirements map poorly onto Level 1 systems.

Effective AI governance must be designed for Level 1 entities: technical constraints (make undesired behavior technically impossible), continuous monitoring (distributional shift, optimization drift, goal misgeneralization), automatic shutdowns, and strict liability for deployers. These principles align with the enforcement mechanisms proposed here: output-based monitoring, automatic consequences, verification without requiring agreement.

V.B. For Legal Reform

Against Normative Inflation. Normative inflation occurs when legal systems create more rules faster than they can enforce. Adding more rules without increasing enforcement resources spreads enforcement thinner, reducing compliance with all rules. Each new rule should be evaluated on enforceability grounds. Rules that cannot be enforced algorithmically and lack sufficient human enforcement resources should not be enacted.

Against Normative Deflation. Normative deflation occurs when legal systems make rules very difficult to change. Under heterogeneity, normative deflation creates arbitrage opportunities. Level 1 entities optimize against static rules over time. The solution is institutional flexibility: rules must be updatable when gaming is detected or when circumstances change.

Some jurisdictions exhibit legal stagflation: simultaneous normative inflation (too many new rules, weakly enforced) and normative deflation (core constitutional rules frozen, cannot adapt). Argentina exemplifies this pattern with a rigid constitution (CLI 0.89) combined with thousands of selectively enforced regulatory rules (Lerer, 2025h). Escaping stagflation requires reducing unenforceable rules, increasing algorithmic enforcement, and reducing constitutional rigidity for rapidly-changing domains.

VI. LIMITATIONS AND FUTURE RESEARCH

VI.A. Limitations of This Analysis

Empirical limitations. The evidence presented is illustrative, not systematic. A definitive test of the theory would require controlled comparisons: jurisdictions with algorithmic enforcement versus traditional enforcement, holding constant other variables. Such studies do not yet exist.

Measurement challenges. The framework posits that entities differ in intentionality level, but I have not provided a validated measurement instrument. Corporate form is used as a crude proxy. Without a reliable way to measure intentionality level, testing the theory's predictions is difficult.

Normative limitations. The paper focuses primarily on effectiveness: which enforcement mechanisms produce compliance? It does not fully develop normative questions: which enforcement mechanisms are just? Algorithmic enforcement may be effective but normatively troubling.

Implementation challenges. The gap between theoretical concept and practical realization may be larger than this paper suggests. Designing algorithmic enforcement systems is technically complex, politically contentious, and expensive.

VI.B. Future Research Directions

The framework suggests several promising research directions: measuring intentionality-robustness through operationalizing intentionality level and testing mechanism performance; comparative enforcement studies identifying jurisdictions with algorithmic versus traditional

enforcement; optimal enforcement mixes under varying domain characteristics; mechanism design under heterogeneity extending theory to explicitly incorporate heterogeneous intentionality; and normative theory of enforcement under heterogeneity developing principles for evaluating enforcement mechanisms when targets differ in cognitive capacities.

VII. CONCLUSION

Legal systems invest enormous resources in enforcement, yet compliance remains stubbornly low and recidivism disturbingly high, particularly among corporate entities. This paper has argued that the enforcement puzzle reflects an architectural mismatch: mechanisms designed for human moral agents (Level 3 intentionality) fail when applied to corporate optimizers (Level 1 intentionality).

Drawing on Asymmetric Intentionality Theory and the Normative Calculation Problem, I argued that this mismatch is not merely an implementation failure (wrong sanction levels, insufficient detection) but a structural failure (wrong type of mechanism for the type of target). Traditional enforcement relies on capacities Level 1 entities lack: forming common knowledge, experiencing shame, valuing reputation intrinsically, understanding spirit behind letter, internalizing principles, engaging in moral learning. When enforcement presupposes these capacities, it fails systematically.

The evidence across domains supports this diagnosis. Corporate criminal liability generates persistent recidivism despite massive penalties because the enforcement mechanism presumes mens rea and rehabilitation, neither of which applies to Level 1 optimizers. Environmental enforcement fails because it presumes stewardship and intrinsic environmental valuation, capacities corporations lack. The pattern is consistent: Level 3 enforcement mechanisms applied to Level 1 entities produce compliance theater, letter-versus-spirit exploitation, and persistent recidivism.

The solution I have proposed is enforcement mechanisms that do not rely on Level 3 capacities: algorithmic enforcement operating on verifiable outputs, automatic consequences, bright-line rules, verification without requiring agreement or common knowledge. Such mechanisms are intentionality-robust: they work regardless of whether the target is a moral agent or an optimizer.

I have been careful to acknowledge that algorithmic enforcement is not a panacea. It faces significant limitations: Goodhart's Law, capture risk, rigidity, and the irreducible ambiguity problem. Hybrid systems combining algorithmic verification for measurable outputs with human judgment for ambiguous standards are likely optimal in most domains.

The analysis has implications beyond corporate enforcement. Mechanism design must be revised to incorporate intentionality-robustness as a design criterion. AI governance must be designed for Level 1 systems rather than anthropomorphizing artificial optimizers. Legal reform must resist both normative inflation and normative deflation.

Ludwig von Mises demonstrated that economic coordination requires prices because prices aggregate information dispersed across countless actors. This paper has argued that legal coordination faces an analogous problem, aggravated when actors process legal signals in structurally different ways. When Level 3 agents and Level 1 entities read the same precedent and extract different information, the pricing mechanism breaks down. Legal prices bifurcate, intentionality arbitrage becomes possible, and enforcement fails.

The solution is not more enforcement of the same type. It is different enforcement: mechanisms that generate signals readable by all actors regardless of intentionality level. Algorithmic enforcement, with all its limitations and risks, at least has this virtue: it does not require the target to be moral. It works for optimizers and for agents, for humans and for algorithms, for Level 1 and Level 3. In a world where heterogeneous intentionality is unavoidable, enforcement mechanisms must be designed for the actors they actually regulate, not the actors legal theory imagines.

The path forward is not simple. Implementing intentionality-robust enforcement requires technical capacity, political will, and institutional redesign. It requires confronting uncomfortable truths: that corporations are not moral persons in any sense relevant to criminal liability, that voluntary compliance programs are often theater, that exhorting entities to "act ethically" is category error. But confronting these truths is preferable to continuing to invest billions in enforcement mechanisms that predictably fail.

ACKNOWLEDGMENTS AND AI DISCLOSURE

This paper synthesizes and extends research developed across a series of working papers on Asymmetric Intentionality Theory. The author gratefully acknowledges productive engagement with AI systems (Claude, Anthropic) in the mathematical formalization of theoretical constructs and in organizing the integration of multiple research streams. All theoretical frameworks, legal interpretations, empirical observations, and errors remain the sole responsibility of the author.

REFERENCES

Aumann, R. J. (1976). Agreeing to disagree. *Annals of Statistics*, 4(6), 1236-1239.

Becker, G. S. (1968). Crime and punishment: An economic approach. *Journal of Political Economy*, 76(2), 169-217.

CFPB (2016). Consumer Financial Protection Bureau fines Wells Fargo $100 million for widespread illegal practice of secretly opening unauthorized accounts. Press Release, September 8, 2016.

CFPB (2022). Consumer Financial Protection Bureau takes action against Wells Fargo for illegal surprise overdraft fees. Press Release, December 20, 2022.

CNBC (2016). Volkswagen memos suggest emissions problem was known earlier. News Report, February 18, 2016.

Dennett, D. C. (1987). *The Intentional Stance*. Cambridge, MA: MIT Press.

Garrett, B. L. (2014). *Too Big to Jail: How Prosecutors Compromise with Corporations*. Cambridge, MA: Harvard University Press.

Goodhart, C. A. E. (1975). Problems of monetary management: The U.K. experience. *Papers in Monetary Economics*, Vol. I, Reserve Bank of Australia.

Gray, W. B., & Shimshack, J. P. (2011). The effectiveness of environmental monitoring and enforcement: A review of the empirical evidence. *Review of Environmental Economics and Policy*, 5(1), 3-24.

Hayek, F. A. (1945). The use of knowledge in society. *American Economic Review*, 35(4), 519-530.

Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2), 263-291.

Laibson, D. (1997). Golden eggs and hyperbolic discounting. *Quarterly Journal of Economics*, 112(2), 443-477.

Lerer, I. A. (2025a). Asymmetric Intentionality in Law and Game Theory: When Legal Systems Assume Moral Agency That Doesn't Exist. SSRN Working Paper.

Lerer, I. A. (2025b). The Dennett-Nash Equilibrium: Game Theory Under Heterogeneous Intentionality. SSRN Working Paper.

Lerer, I. A. (2025c). The Aumann Impossibility Theorem Under Heterogeneous Intentionality: When Rational Agents Can Agree to Disagree. SSRN Working Paper.

Lerer, I. A. (2025d). The Generalized Intentionality Mismatch Theorem: When Law Assumes Moral Agency That Doesn't Exist. SSRN Working Paper.

Lerer, I. A. (2025e). Intentionality Mismatch in Legal Systems: A General Framework. SSRN Working Paper.

Lerer, I. A. (2025f). Corporate Criminal Liability and the Impossibility of Mens Rea: Why Guilty Minds Require Moral Agents. SSRN Working Paper.

Lerer, I. A. (2025g). Environmental Law and the Myth of Corporate Citizenship: Intentionality Mismatch in Ecological Regulation. SSRN Working Paper.

Lerer, I. A. (2025h). Legal Prices: The Normative Calculation Problem and Why Legal Systems Need Price Mechanisms. SSRN Working Paper (under review).

Mises, L. von (1920). Die Wirtschaftsrechnung im sozialistischen Gemeinwesen. *Archiv für Sozialwissenschaft und Sozialpolitik*, 47, 86-121.

NPR (2015). VW engineers warned of emissions problem, Congressional testimony. News Report, October 8, 2015.

OCC (2021). Office of the Comptroller of the Currency assesses $250 million civil money penalty against Wells Fargo Bank. Press Release, September 9, 2021.

Potoski, M., & Prakash, A. (2005). Green clubs and voluntary governance: ISO 14001 and firms' regulatory compliance. *American Journal of Political Science*, 49(2), 235-248.

Thaler, R. H., & Sunstein, C. R. (2008). *Nudge: Improving Decisions About Health, Wealth, and Happiness*. New Haven, CT: Yale University Press.

U.S. Sentencing Commission (2021). *Recidivism of Federal Offenders Released in 2010*. Washington, D.C.